

تدریس ریاضیات

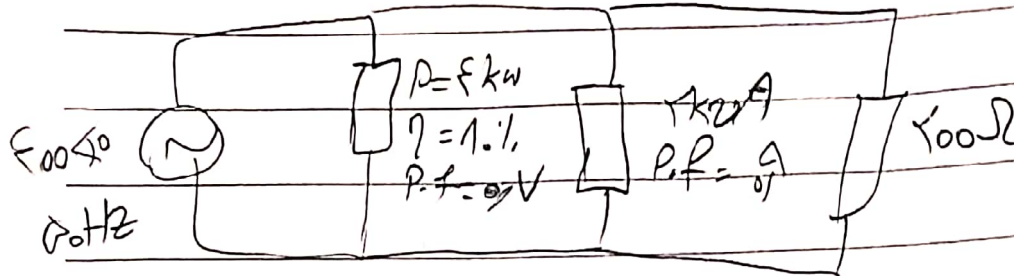
ارسی بزرگ ۱۴۰۱/۰۲/۰۵

Date: / /

Sat Sun Mon Tue Wed Fri

Subject: _____

سوال ۲: (الف)



$$P_i = \frac{P_{out}}{\eta} = \frac{4}{0.9} = 4.44 \text{ kW}$$

ب. راول

$$Q_1 = P_i \tan(\cos^{-1}(0.9)) = 4.44 \times 0.484 = 2.15 \text{ kVAR (سلفر)}$$

بار دوم:

$$P_r = S \times P.F. = 4.44 \times 0.9 = 3.99 \text{ kW}$$

$$Q_r = -S \sin(\cos^{-1}(0.9)) = -4.44 \times 0.436 = -1.94 \text{ kVAR (کپاسیتر)}$$

بار سوم:

$$V_{ph} = \frac{400}{\sqrt{3}} = 230.94 \text{ V}$$

$$P_r = 3 \times \frac{V_{ph}^2}{R} = 3 \times \frac{230.94^2}{200} = 0.79 \text{ kW}$$

$$Q_r = 0 \text{ kVAR (مقاومت)}$$

کل بار:

$$P_{total} = 4.44 + 3.99 + 0.79 = 9.22 \text{ kW}$$

$$Q_{total} = 2.15 + (-1.94) + 0 = 0.21 \text{ kVAR}$$

$$S_{total} = \sqrt{P_{total}^2 + Q_{total}^2} = \sqrt{9.22^2 + 0.21^2} = 9.23 \text{ kVA}$$

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Date:

☐ Sun ☐ Mon ☐ Tue ☐ Wed ☐ Thu ☐ Fri ☐ Sat

Subject:

$$PF_{total} = \frac{P_{tot}}{S_{tot}} = \frac{V,4}{1,492} \approx 0,94 \text{ ; } \text{بفاز}$$

$$I_L = \frac{S_{tot}}{\sqrt{3} V_L} = \frac{1492}{\sqrt{3} \times 400} \approx 12,108 \text{ A}$$

$$Q_{new} = P_{tot} \tan(\cos^{-1}(0,94)) = V,4 \times 0,992 = 4 \text{ kVAR} \quad (ب)$$

$$Q_C = Q_{tot} - Q_{new} = 5,44 - 4 = 1,44 \text{ kVAR}$$

$$Q_C = \sqrt{3} \times V_{ph} \times \omega C \Rightarrow C = \frac{Q_C}{\sqrt{3} \times V_{ph} \times \omega} = \frac{1440}{\sqrt{3} \times (400 \times 0,94) \times 2\pi \times 50} \approx 1,44 \mu F$$

ظرفیت برق زائد حذف شود

$$S_{new} = \frac{P_{tot}}{PF_{new}} = \frac{V,4}{0,94} \approx 1,105 \text{ kVA}$$

$$I_{L,new} = \frac{1,105}{\sqrt{3} \times 400} \approx 11,1 \text{ A}$$

$$Q_C = Q_{tot} = 5,44 \text{ kVAR} \Rightarrow C = \frac{5440}{\sqrt{3} \times 400 \times 2\pi \times 50} \approx 1,44 \mu F \quad (ج)$$

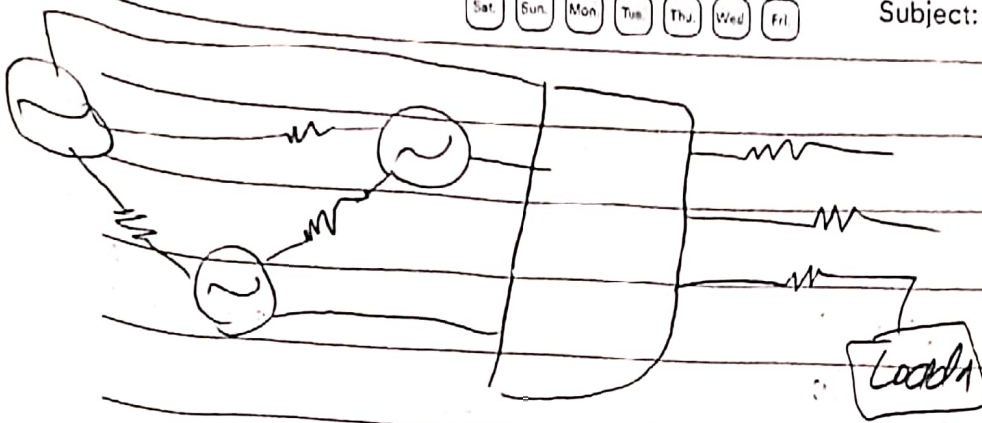
ظرفیت برق زائد حذف شود

$$S_{new} = P_{tot} = V,4 \text{ kVA} \Rightarrow I_{L,new} = \frac{V,400}{\sqrt{3} \times 400} \approx 10,17 \text{ A}$$

Date: / /

Sat Sun Mon Tue Wed Fri

Subject:



داده

الف

$$V_{L, RMS} = 94.0 \text{ V} \rightarrow V_{PH} = \frac{0.707}{\sqrt{2}} \times 94.0 \text{ V} = 61.95 \text{ V}$$

$$Z_g = 0.01 + j0.15 \rightarrow Z_g = \frac{Z_g}{\sqrt{2}} = 0.005 + j0.08 \Omega$$

$$Z_f = 0.02 + j0.5 \Omega$$

$$Z_L = 10 + j10 \Omega$$

$$Z_{LD} = 9 - j10 \Rightarrow Z_L = \frac{Z_{LD}}{\sqrt{2}} = 5 - j10 \Omega$$

$$Z_P = Z_L \parallel Z_L = \frac{(10 + j10)(5 - j10)}{(10 + j10) + (5 - j10)} = \frac{140 - j10}{15} = 9.33 - j6.67 \Omega$$

$$\Rightarrow Z_P \approx 10.154 \angle -34.91^\circ \Omega$$

$$\Rightarrow I_L = \frac{V_{PH}}{Z_{total}} \Rightarrow Z_{total} = Z_g + Z_f + Z_P = 0.005 + j0.08 \Omega$$

$$* 0.005 + j0.08 \Omega + 10.154 - j9.44 \Omega = 10.159 \angle -5.18^\circ \Rightarrow I_L = \frac{61.95 \angle 0^\circ}{10.159 \angle -5.18^\circ} = 6.09 \angle 5.18^\circ \text{ A}$$

جواب

Date:

Subject:

$$V_{\text{Load ph}} = \bar{I}_L \times Z_p = (21.1 \angle -45.1^\circ) \times (101.624 \angle -99.4^\circ) = 2149.43 \angle -144.5^\circ$$

التيار في الحمل : $V_{\text{Load Q}} = 2149.43 \times \sqrt{2} = 3051.15 \text{ V}$

التيار في الحمل : $V_{\text{term, ph}} = V_{\text{Load ph}} + \bar{I}_L Z_f = 2149.43 \angle -144.5^\circ + (21.1 \angle -45.1^\circ) \times (101.624 \angle -99.4^\circ)$

$$\Rightarrow V_{\text{term, ph}} = 2149.43 \angle -144.5^\circ + (21.1 \angle -45.1^\circ) \times (101.624 \angle -99.4^\circ)$$

$$= 2149.43 \angle -144.5^\circ + 2149.43 \angle -144.5^\circ = 4298.86 \angle -144.5^\circ$$

$$\Rightarrow V_{\text{term, L}} = 4298.86 \times \sqrt{2} = 6051.15 \text{ V}$$

$$\bar{I}_L = 21.1 \angle -45.1^\circ \text{ A}$$

التيار في الحمل : $\bar{I}_{\text{ph, S}} = \frac{\bar{I}_L}{\sqrt{3}} \angle (\theta_{\bar{I}_L} + 30^\circ) = \frac{21.1 \angle -45.1^\circ}{\sqrt{3}} \angle 14.9^\circ$

$$\Rightarrow \bar{I}_{L1} = \frac{V_{\text{Load, ph}}}{Z_{L1}} = \frac{2149.43 \angle -144.5^\circ}{101.624 \angle -99.4^\circ} = 21.1 \angle -45.1^\circ \text{ A}$$

$$\Rightarrow \bar{I}_{\text{ph, S}} = \frac{\bar{I}_L}{\sqrt{3}} \angle (\theta_{\bar{I}_L} + 30^\circ) = \frac{21.1 \angle -45.1^\circ}{\sqrt{3}} \angle 14.9^\circ = 12.1 \angle 14.9^\circ \text{ A}$$

Date:

Sat Sun Mon Tue Wed Fri

Subject: _____

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$$\theta_{V-I} = -1/11^\circ - 24/18^\circ = -34,91^\circ \Rightarrow PF_{\text{load}} = \cos(-34,91^\circ)$$

بسیار خوب
0,177

ضرب
در
هم

$$\theta_{V-I} = -0/11^\circ - 24/18^\circ = -34,91^\circ \Rightarrow PF_{\text{term}} = \cos(-34,91^\circ)$$

بسیار خوب
0,177

$$P(t) = V(t)I(t)$$

و این را

$$\begin{cases} V(t) = V_m \cos(\omega_0 t) \\ I(t) = I_m \cos(\omega_0 t + \varphi) \end{cases} \Rightarrow P(t) = V_m I_m \left(\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \right)$$

$$\left(\frac{e^{j(\omega_0 t + \varphi)} + e^{-j(\omega_0 t + \varphi)}}{2} \right)$$

$$= \frac{V_m I_m}{2} \left(e^{j(\omega_0 t + \varphi)} + e^{-j(\omega_0 t + \varphi)} \right)$$

$$= \frac{V_m I_m}{2} \left(\cos(\varphi) + \cos(\omega_0 t + \varphi) \right) = \frac{V_m I_m}{2} \cos(\varphi)$$

$$\cos \varphi = \frac{P}{V_m I_m} \rightarrow \cos \varphi = 0,177 \rightarrow \varphi = \cos^{-1}(0,177) = 90^\circ$$

$\varphi = -90^\circ$ - بسیار خوب

Date: / /

Sat Sun Mon Tue Wed Thu Fri

Subject:

$$\vec{V} = \angle 21.4^\circ$$

$$\vec{I} = \angle 1.214^\circ \rightarrow r_{ms} = 22.0^\circ$$

$$\vec{I} = \angle 1.214^\circ \rightarrow r_{ms} = 18^\circ \angle 4^\circ$$

$$r_{AR} = 291V = \frac{\sqrt{3}}{2} \times 465 = 172.5 \text{ W}$$

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$$Z_f = \angle \times (0.105 + j0.1) = 0.105 + j0.1 \Omega$$

$$Z_{L1} = 10 + j12 \Omega$$

$$S_{cp} = \frac{5000 + j10000}{2} = 1250 + j5000 \text{ VA}$$

$$I_L = \frac{V_s}{Z_f + Z_{L1}} = \frac{22.0 \angle 19.5^\circ}{(0.105 + j0.1) + (10 + j12)} = \frac{22.0 \angle 19.5^\circ}{10.105 + j12.1}$$

$$V_{L1} = I_L \times Z_{L1} = 12.10 \angle -0.9^\circ \times (10 + j12) = 22.4 \angle 11.2^\circ \text{ V}$$

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Date:

Sat Sun Mon Tue Wed Thu Fri

Subject:

$$V_L^{(k+1)} = V_S - Z_{fx} \left(\frac{V_L^{(k)}}{Z_L} + \frac{S_{Lx}^*}{Z_L^{(k)*}} \right) \quad \text{حالت دوم}$$

تکرار اول (k=0) با فرض ولتاژ شروع 0 و 0.95 A

$$I_{total} = \frac{230.95}{10 + j0.5} + \frac{15000 - j1000}{250.95} = (4.1 - j0.44) + (0.17 - j0.004) = 4.27$$

$$V_L^{(1)} = 250.95 - (0.12 + j0.003) \times 4.27 \times 1000 = 248.84 - j0.27$$

$$V_L \rightarrow V_{LL} = 248.84 + j0.119 \quad \text{تکرار دوم (k=1)}$$

(ب) ضریب تغییرات ولتاژ

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